Resultater fra QH2 for Hilde Celine Løvland

Resultat for dette forsøket: 2 av 9
Innlevert 3. sep. i 16.22
Dette forsøket tok 25 minutter

Riktig svar

Spørsmål 1
1 / 1 poeng

Hydraulic cylinder, ideal, basic flow/speed relations . The cylinder is extracting . Cylinder diameter (d) = 200 mm . Cylinder rod diameter (d_R) = 100 mm . Cylinder out-flow (Q_o) = 105 l/min . What is the cylinder in-flow (Q_i) in l/min?

140

140 (med marginen: 1,4)

Feil svar

Spørsmål 2
0 / 1 poeng

Hydraulic cylinder, ideal, basic pressure/force relations . The cylinder is extracting . The cylinder is subjected to an assistive (negative) load . Cylinder diameter (d) = 250 mm . Cylinder rod diameter (d_R) = 150 mm . Cylinder rod side pressure (p_2) = 183 bar . Cylinder piston side pressure (p_1) = 10 bar . What is the cylinder load force (F_L) in kN?

543,4955

525,8241 (med marginen: 5,2582)

Riktig svar

Spørsmål 3
1 / 1 poeng

Hydraulic cylinder, basic pressure/force relations . The cylinder is retracting . The cylinder is subjected to an assistive (negative) load . Cylinder diameter (d) = 80 mm . Cylinder rod diameter (d_R) = 50 mm . Cylinder piston side pressure (p_1) = 52 bar . Cylinder hydromechanical efficiency (eta_hmC) = 0.87 . Cylinder load force (F_L) = 22.8 kN . What is the cylinder rod side pressure (p_2) in bar?

19,4612

19,4612 (med marginen: 0,1946)

Ubesvart

Spørsmål 4
0 / 1 poeng

Hydraulic cylinder, buckling, basic relations . End mounted, steel rod . Cylinder rod diameter (d_R) = 100 mm . Cylinder base length (L_1) = 1974 mm . Cylinder total length (L) = 2785 mm . What is the cylinder buckling force (F_bck) in kN?

4 504,4616 (med marginen: 45,0446)

Ubesvart

Spørsmål 5
0 / 1 poeng

Hydraulic cylinder, buckling, design considerations . Foot mounted, steel rod . Cylinder base length (L_1) = 0 mm . Cylinder total length (L) = 122 mm . Safety against buckling (S) = 6 . Cylinder load force (F_L) = 1450.1 kN . What is the minimum cylinder rod diameter (d_R) in mm?

33,5888 (med marginen: 0,3359)

Ubesvart

Spørsmål 6
0 / 1 poeng

Accumulator, basic relations . Isothermal process, 1 -> 2 . Final accumulator volume (V_2) = 21.2 l . Initial accumulator pressure (p_1) = 157 bar . Final accumulator pressure (p_2) = 49 bar . What is the initial accumulator gas volume (V_1) in l?

6,6166 (med marginen: 0,0661)

Ubesvart

Spørsmål 7
0 / 2 poeng

Accumulator, design considerations I . Compute accumulator preload (total) volume . Accumulator polytropic coefficient (n_L) = 1.4 . Change in volume (DeltaV) = 4.7 l . Accumulator minimum load pressure (p_min) = 63 bar . Accumulator maximum load pressure (p_max) = 119 bar . Accumulator load pressure (p_L) = 69 bar . Accumulator preload pressure (p_0) = 57 bar . What is the preload (total) accumulator gas volume (V_0) in l?

14,6032 (med marginen: 0,146)

Ubesvart

Spørsmål 8

0 / 1 poeng

Accumulator, design considerations II . Accumulator connected to ideal motor . Motor compressing accumulator in adiabatic process . Motor displacement (D) = 500 cm^3/rev . Motor revolutions during process (Deltal_M) = 11 rev . Initial accumulator pressure (p_1) = 51 bar . Final accumulator pressure (p_2) = 67 bar . Accumulator polytropic coefficient (n_L) = 1.4 . What is the final accumulator gas volume (V_2) in l?

25,5582 (med marginen: 0,2555)

Quizresultat: 2 av 9